

Tatheer Aima Naqvi

CS student Nust Islamabad



PROFESSIONAL SUMMARY

First-year CS student at NUST with an unusual background — I spent my pre-university years doing serious physics, placing 7th nationally out of 7000 students, training in nuclear labs at PIEAS, and proposing beamline experiments for CERN. I came into CS already knowing how to think rigorously about hard problems. Since starting, I've deployed a YOLOv8-based AI surveillance system in production, built projects across C++, Java, and React, and completed CS50's AI curriculum. Outside technical work, I founded UMMED to expand STEM access for rural women in Pakistan.

CONTACT

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Work Experience

Backend Developer Intern

ZEMT Innovations - NUST CEC Lab

Designed the overall system architecture and led the integration of a YOLOv8-based machine learning model for the Urban Guard project. Developed backend components for real-time CCTV analytics, enabling automated object detection and surveillance monitoring. Contributed to the successful deployment of a real-world AI-powered security solution from development to implementation.

Social Media Marketing Intern

Echolens

Remote internship

Managed the company's LinkedIn platform and established its Instagram presence from the ground up. Created and curated content, supported digital marketing campaigns, and enhanced brand visibility through strategic social media management, creativity, and consistent audience engagement.

Founder - Umeed Organization

UMMED – Rural Women in STEM Initiative

Founded UMMED, an initiative dedicated to empowering rural women through STEM education and mentorship. Led efforts to promote educational opportunities, digital literacy, and skill development, helping young women gain the knowledge and confidence needed to pursue careers in science, technology, engineering, and mathematics. Advocated for greater accessibility and inclusion in education, working to bridge the gap between rural communities and STEM opportunities.

SKILLS

- **Full-stack development (Java)** Intermediate
- **C++ programming with SFML and raylib library** - Intermediate
- **Experimental design** Intermediate
- **Scientific programming Monte Carlo simulation** Intermediate
- **Mathematical modeling & research methodology** Advanced
- **Statistical analysis and particle physics** Advanced
- **Public Speaking (2x national winner)** Expert

EDUCATION

BS CSNUST UNIVERSITY ISLAMABAD

Sep 2025 till present

- Second Semester Student Nust
- CGPA : 3.41
- Executive research , NPAS society NUST
- Executive Marketing ,Google Developer Club

Fsc - ChenabCollegeJhang

Aug 2023 - Aug 2025

- FSC - 1118/1200
- Matric 1080 / 1100
- Head Girl - led student body
- President Science Society
- President Debating Society

PROJECTS

- **AI Projects Implementation (CS50's Introduction to Artificial Intelligence with Python)**

Developed and implemented a collection of AI projects covering search algorithms, knowledge representation, probabilistic reasoning, optimization, machine learning, and natural language processing. Applied concepts including PageRank, Minimax, Constraint Satisfaction Problems (CSPs), Bayesian Networks, and Neural Networks, demonstrating practical experience in building intelligent systems and solving real-world computational problems.

https://github.com/aima0749/Ai_assignments_CS50

- **Finance Tracker**

Console-based personal finance tracker built with Java, applying object-oriented programming principles

<https://github.com/hashimkyn/Finance-Tracker>

- **Physics Sand-box**

Interactive 2D physics simulation for visualizing and learning collision mechanics using C++ and raylib library.

<https://github.com/aima0749/PHYSICSENGINE>

- **Zenvy (passion project)**

Full developed application to connect users with local service providers, built with React JS and Supabase backend.

<https://github.com/aima0749/Zenvy>

- **Bank Management System**

OOP-based bank management application with graphical interface built using the SFML library in C++.

<https://github.com/aima0749/Banking-management-System->

Awards

Bronze Medalist , International Nuclear Science Olympiad, Malaysia

International Atomic Energy Commission

Demonstrated advanced expertise in nuclear science through problem-solving and experimental analysis, earning a bronze medal at an international level with particular proficiency in radiation physics, nuclear reactors, and safety protocols. Completed 200+ hours of hands-on training in nuclear labs at PIEAS and trained by top Nuclear Scientists of Pakistan and worked in research reactor PARR.

Beamline CERN

CERN

Proposed and optimized 2 high-precision particle beamline experiments for CERN, enhancing experimental accuracy and contributing to groundbreaking physics research outcomes after simulating it with Monte-Carlo simulation.Part of a team of 6 people where we proposed two experiments on neutron spallation and radio-technologies.

National Physics Talent Contest

HEC Islamabad

Achieved 7th rank out of 7000 students nation-wide in Physics Talent Contest and was selected and honored to be part of reserved team of Pakistan, representing Pakistan its International Physics Olympiad after attending three physical camps in PIEAS.

National Debate

National Debater Pakistan National Debate Circuit

Represented my institution as a two-time national debater, demonstrating excellence in public speaking, persuasive communication, critical thinking, and argumentation. Competed against top students from across the country while developing strong leadership and analytical skills.

LANGUAGES

- **Urdu** - Native
 - **Punjabi** - Native
 - **Hindko** - Native
 - **English** - Fluent
 - **Arabic** - Intermediate
 - **Hindi** - Conversational
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